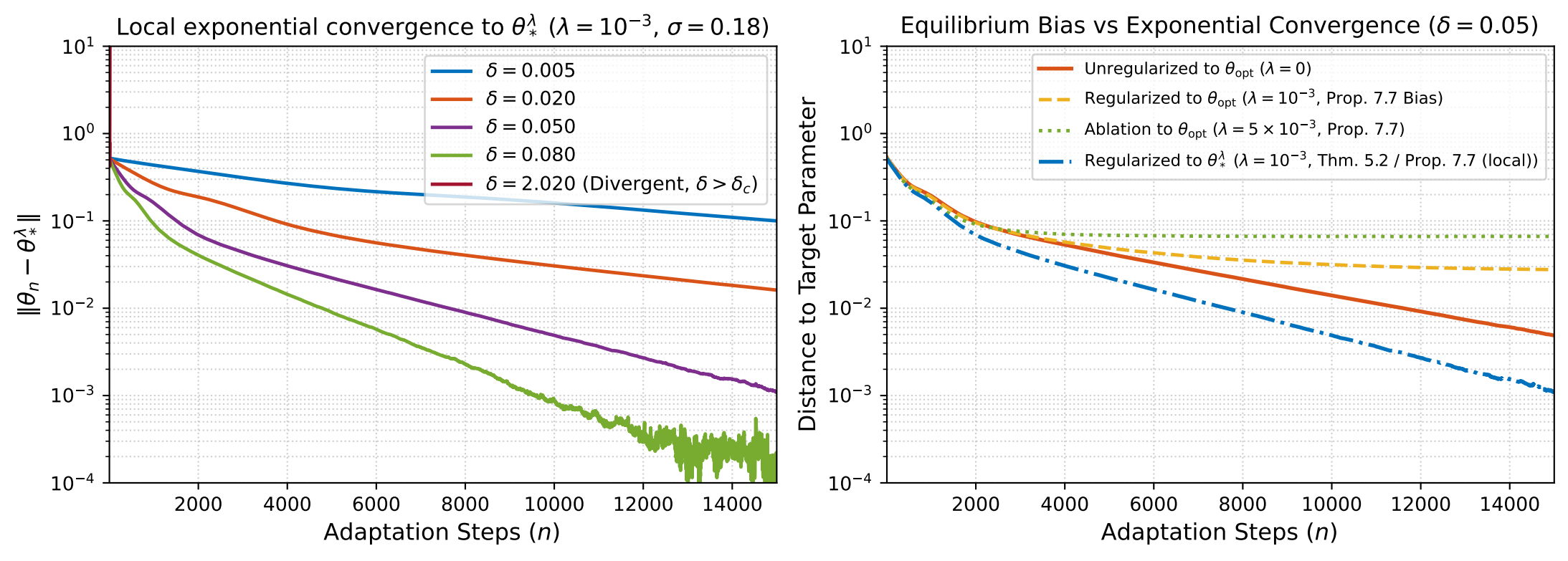

Local exponential convergence to θ_*^λ ($\lambda = 10^{-3}$, $\sigma = 0.18$)
$\|\theta_n - \theta_*^\lambda\|$
Adaptation Steps (n)
$\delta = 0.005$
$\delta = 0.020$
$\delta = 0.050$
$\delta = 0.080$
$\delta = 2.020$ (Divergent, $\delta > \delta_c$)
Equilibrium Bias vs Exponential Convergence ($\delta = 0.05$)
Distance to Target Parameter
Adaptation Steps (n)
Unregularized to θ_{opt} ($\lambda = 0$)
Regularized to θ_{opt} ($\lambda = 10^{-3}$, Prop. 7.7 Bias)
Ablation to θ_{opt} ($\lambda = 5 \times 10^{-3}$, Prop. 7.7)
Regularized to θ_*^λ ($\lambda = 10^{-3}$, Thm. 5.2 / Prop. 7.7 (local))